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Two Dimensional Successive Transformations in Computer Graphics: Mathematical and Matrix Notation

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Abstract: Computer graphics has vital role to the advancement of devices and technology. In computer graphics, transformations are used to repositioning, changing orientation, alteration of 2D object, mirror image and slide over. There are some 2D transformation operations which are translation, rotation, scaling, reflection, shearing and some composite transformation along with successive transformation. This paper discussed a comprehensive overview of composition of transformations or successive transformations. Moreover, the mathematical and matrix representation of each successive transformation operations are also mentioned. Mathematical and matrix notation can be helpful to give the accurate result to computer scientists in graphic evaluation through modification.

Keywords: Translation, Rotation, Scaling, Composite transformation, 2D Successive

INTRODUCTION

In computer graphics, when any object or image depicted with X and Y axis known as two dimensional (2D) object or image. Where that object or image represented known as 2D space. Multiple operations can be performed on the object. These operations mean mapping or transformation that modifying the different parameters of object. If this transformation applies on 2D object then it is called 2D transformation. Under 2D transformation many operations are there such as translation, rotation, scaling, reflection and shearing [1][2] shown in Fig. 1. These all operations have own parameters that alter the size or location of object[3]. As translation means repositioning the object, rotation is implementing for orientation (clockwise or anti-clockwise) scaling is used for alteration in size, reflection is for mirror image and shearing for slide over. Individual operation can be implemented on 2D object, but two operations (same or different) can also be implemented on object then it is known as composite transformation. Here composite transformation is also having various types such as successive-translation, successive-rotation, successive-scaling, general pivot point rotation and general pivot point scaling. But here this paper discusses about the 2D successive transformation operations shown in Fig. 2.

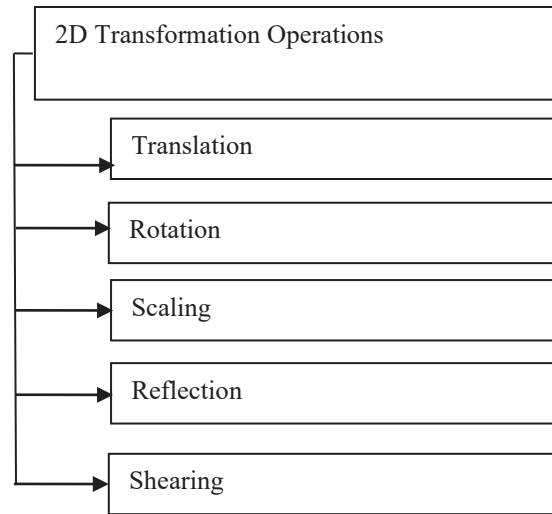


FIGURE 1.2D Transformation Operations

Each operation can also be represented by matrix format[4] which is a mathematical tool. This gives easy and accurate representation of transformation values of the 2D object of before and after transformation. By matrix operations calculation can be easily and effectively[5].

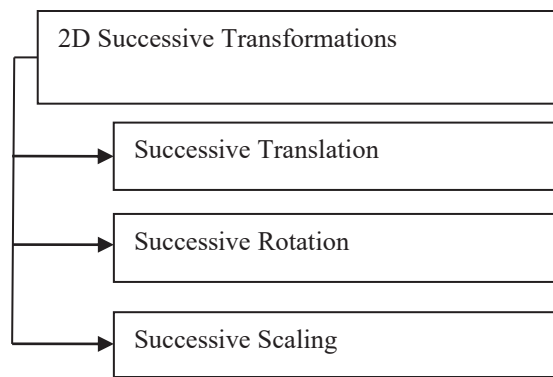


FIGURE 2.2D Successive Transformations

REVIEWING 2D SUCCESSIVE TRANSFORMATIONS

Two dimensional transformations[6][7] can be composite or combine[8] when multiple operations perform together. Composite transformation can be achieved by matrix multiplication. For example X and Y both are represented as matrices of any two types of transformation[9] want to apply on 2D object. Initially apply the first transformation i.e X and later Y transformation, can be multiply each other. Finally resultant matrix multiplies with 2D object to get the transformed co-ordinates of object.

Types of Composite transformation:-

- Successive translation
- Successive rotation
- Successive scaling
- General pivot point rotation
- General pivot point scaling

As explanation of each type, successive means one by one same operation so in successive translation one by one translation operation takes place and same for rotation and scaling also. Other types of

composite transformation[10] are general pivot point rotation and scaling as well also named as general fixed point. In which one point of 2D object is fixed (pivot) and other co-ordinates will change as per operation. This paper focused on successive transformation only. Translation, rotation and scaling are used on 2D object but at a time one component is manipulated[11].

MATHEMATICAL AND MATRIX REPRESENTATION OF 2D SUCCESSIVE TRANSFORMATIONS

Successive Translation

When two successive translation factors: $(t_{x'}, t_{y'})$ and $(t_{x''}, t_{y''})$ apply to a 2D object one by one known as successive translation[12]. As shown in Fig. 3 initially point P is at original position of object with co-ordinate values (x, y) . After first translation T' the point P is translated to P' position and later on next translation T'' takes place and then point P' translated to the P'' position. Fig 4 shown the sequence of successive translation, initially point P , intermediate point P' and finally point P'' .

Translation factors: $T' = (t_{x'}, t_{y'})$ and $T'' = (t_{x''}, t_{y''})$

Point: $= (x, y)$, $P' = (x', y')$ and $P'' = (x'', y'')$

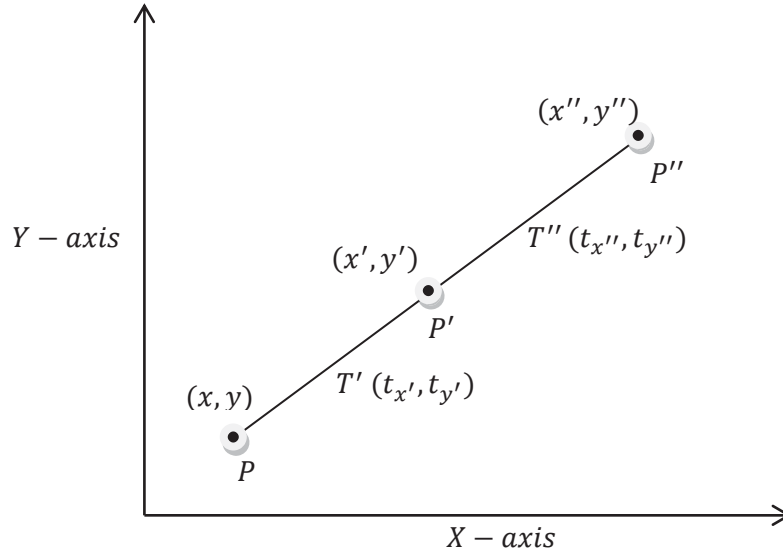


FIGURE 3. Successive translation

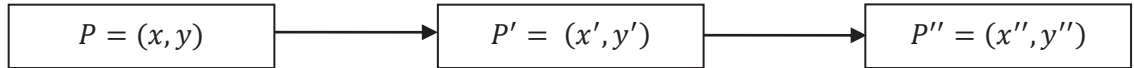


FIGURE 4. Sequence of successive translation

Mathematical Equations and Matrix

Equation 1 and 2 show the mathematical notation of successive translation. The transformation is translated by sequence from $(t_{x'}, t_{y'})$ to $(t_{x''}, t_{y''})$.

Initial translation

$$P' = T'.P \quad (1)$$

$$P'(x', y') = T'(t_{x'}, t_{y'}).P(x, y)$$

$$\text{where } x' = x + t_{x'}, y' = y + t_{y'}$$

Matrix representation for translation mentioned below

$$\begin{bmatrix} 1 & 0 & t_{x'} \\ 0 & 1 & t_{y'} \\ 0 & 0 & 1 \end{bmatrix}$$

Next translation:

$$P'' = T'' \cdot P' \quad (2)$$

$$P''(x'', y'') = T''(t_{x''}, t_{y''}) \cdot P'(x', y')$$

$$\text{where } x'' = x' + t_{x''}, y'' = y' + t_{y''}$$

Matrix representation for next translation mentioned below

$$\begin{bmatrix} 1 & 0 & t_{x''} \\ 0 & 1 & t_{y''} \\ 0 & 0 & 1 \end{bmatrix}$$

Composition of 2D translation is shown by using equation 3.

$$P'' = T'' \cdot T' \cdot P \quad (3)$$

$$P''(x'', y'') = T''(t_{x''}, t_{y''}) \cdot T'(t_{x'}, t_{y'}) \cdot P(x, y)$$

$$\text{where } x'' = x + t_{x'} + t_{x''}, y'' = y + t_{y'} + t_{y''}$$

The composite transformation matrix for this sequence of translation is

$$T = T''(t_{x''}, t_{y''}) \cdot T'(t_{x'}, t_{y'})$$

$$T = \begin{bmatrix} 1 & 0 & t_{x''} \\ 0 & 1 & t_{y''} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & t_{x'} \\ 0 & 1 & t_{y'} \\ 0 & 0 & 1 \end{bmatrix}$$

$$T = \begin{bmatrix} 1 & 0 & t_{x'} + t_{x''} \\ 0 & 1 & t_{y'} + t_{y''} \\ 0 & 0 & 1 \end{bmatrix}$$

Both $T' = (t_{x'}, t_{y'})$ and $T'' = (t_{x''}, t_{y''})$ translations are equal to the single translation $T = ((t_{x'} + t_{x''}), (t_{y'} + t_{y''}))$. This demonstrates that two successive translations are additive.

Successive Rotation

When two successive rotations applied on object by two angles: β and γ one by one known as successive rotation[12]. As shown in Fig. 4 initially point P is at original position of object with co-ordinate values (x, y) . Apply first rotation by β angle the point P is rotated to P' position and later on next rotation applied by γ and then point P' rotated to the P'' position.

Rotation angles: $R'(\beta)$ and $R''(\gamma)$

Point: $= (x, y)$, $P' = (x', y')$ and $P'' = (x'', y'')$

Mathematical Equations and Matrix

The transformation is applied to a point P produce the transformed position shown in equation 4

$$P'' = R''(\gamma).R'(\beta).P \quad (4)$$

As per fig 5, in ΔOPA equation 5 and 6 represent the value of $\sin(\phi)$ and $\cos(\phi)$

$$\sin(\phi) = \frac{y}{r}, y = r. \sin(\phi) \quad (5)$$

$$\cos(\phi) = \frac{x}{r}, x = r. \cos(\phi) \quad (6)$$

In $\Delta OP'A'$, equation 7 and 8 represents the value of y' and x'

$$\begin{aligned} \sin(\phi + \beta) &= \frac{y'}{r}, y' = r. \sin(\phi + \beta) \\ y' &= r. \sin(\phi). \cos(\beta) + r. \cos(\phi). \sin(\beta) \\ y' &= y. \cos(\beta) + x. \sin(\beta) \end{aligned} \quad (7)$$

$$\begin{aligned} \cos(\phi + \beta) &= \frac{x'}{r}, x' = r. \cos(\phi + \beta) \\ x' &= r. \cos(\phi). \cos(\beta) - r. \sin(\phi). \sin(\beta) \\ x' &= x. \cos(\beta) - y. \sin(\beta) \end{aligned} \quad (8)$$

Matrix representation for rotation with β angle mentioned below

$$R'(\beta) = \begin{bmatrix} \cos(\beta) & -\sin(\beta) & 0 \\ \sin(\beta) & \cos(\beta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

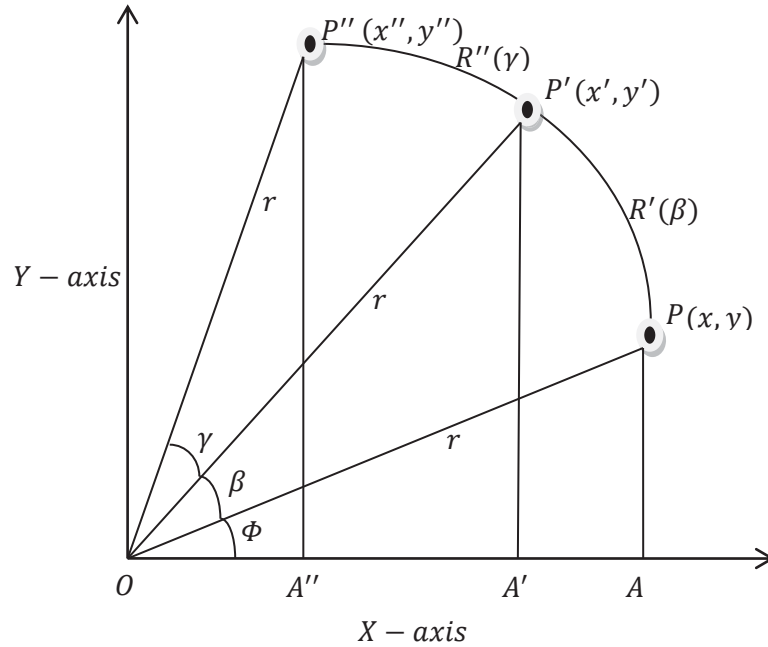


FIGURE 4. Successive rotation

In $\Delta OP''A''$, equation 9 and 10 represent the value of y'' and x''

$$\sin(\phi + (\beta + \gamma)) = \frac{y''}{r}, y'' = r. \sin(\phi + (\beta + \gamma))$$

$$\begin{aligned} y'' &= r. \sin(\phi). \cos(\beta + \gamma) + r. \cos(\phi). \sin(\beta + \gamma) \\ y'' &= r. \sin(\phi). (\cos(\beta). \cos(\gamma) - \sin(\beta). \sin(\gamma)) + r. \cos(\phi). (\sin(\beta). \cos(\gamma) + \cos(\beta). \sin(\gamma)) \end{aligned}$$

$$\begin{aligned}
y'' &= y.(\cos(\beta). \cos(\gamma) - \sin(\beta). \sin(\gamma)) + x.(\sin(\beta). \cos(\gamma) + \cos(\beta). \sin(\gamma)) \\
y'' &= y. \cos(\beta). \cos(\gamma) - y. \sin(\beta). \sin(\gamma) + x. \sin(\beta). \cos(\gamma) + x. \cos(\beta). \sin(\gamma) \\
y'' &= \sin(\gamma). (x. \cos(\beta) - y. \sin(\beta)) + \cos(\gamma). (x. \sin(\beta) + y. \cos(\beta)) \\
y'' &= x'. \sin(\gamma) + y'. \cos(\gamma) \\
(\text{using } x' &= x. \cos(\beta) - y. \sin(\beta), y' = y. \cos(\beta) + x. \sin(\beta)) \\
\cos(\emptyset + (\beta + \gamma)) &= \frac{x''}{r}, x'' = r. \cos(\emptyset + (\beta + \gamma)) \\
x'' &= r. \cos(\emptyset). \cos(\beta + \gamma) - r. \sin(\emptyset). \sin(\beta + \gamma) \\
x'' &= r. \cos(\emptyset). (\cos(\beta). \cos(\gamma) - \sin(\beta). \sin(\gamma)) - r. \sin(\emptyset). (\sin(\beta). \cos(\gamma) + \cos(\beta). \sin(\gamma)) \\
x'' &= x. (\cos(\beta). \cos(\gamma) - \sin(\beta). \sin(\gamma)) - y. (\sin(\beta). \cos(\gamma) + \cos(\beta). \sin(\gamma)) \\
x'' &= x. \cos(\beta). \cos(\gamma) - x. \sin(\beta). \sin(\gamma) - y. \sin(\beta). \cos(\gamma) + y. \cos(\beta). \sin(\gamma) \\
x'' &= \cos(\gamma). (x. \cos(\beta) - y. \sin(\beta)) - \sin(\gamma). (x. \sin(\beta) + y. \cos(\beta)) \\
x'' &= x'. \cos(\gamma) - y'. \sin(\gamma) \\
(\text{using } x' &= x. \cos(\beta) - y. \sin(\beta), y' = y. \cos(\beta) + x. \sin(\beta))
\end{aligned} \tag{9}$$

Matrix representation for rotation with γ angle mentioned below.

$$R''(\gamma) = \begin{bmatrix} \cos(\gamma) & -\sin(\gamma) & 0 \\ \sin(\gamma) & \cos(\gamma) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The composite transformation matrix is shown below by using equation 11.

$$\begin{aligned}
R &= R''(\gamma). R'(\beta) \\
R &= \begin{bmatrix} \cos(\gamma) & -\sin(\gamma) & 0 \\ \sin(\gamma) & \cos(\gamma) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos(\beta) & -\sin(\beta) & 0 \\ \sin(\beta) & \cos(\beta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
R &= \begin{bmatrix} \cos(\gamma). \cos(\beta) - \sin(\gamma). \sin(\beta) & -\cos(\gamma). \sin(\beta) - \sin(\gamma). \cos(\beta) & 0 \\ \sin(\gamma). \cos(\beta) + \cos(\gamma). \sin(\beta) & -\sin(\gamma). \sin(\beta) + \cos(\gamma). \cos(\beta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
R &= \begin{bmatrix} \cos(\gamma + \beta) & -\sin(\gamma + \beta) & 0 \\ \sin(\gamma + \beta) & \cos(\gamma + \beta) & 0 \\ 0 & 0 & 1 \end{bmatrix}
\end{aligned} \tag{11}$$

Both $R'(\beta)$ and $R''(\gamma)$ rotations are equal to the single rotation $R = (\gamma + \beta)$. This demonstrates that two successive rotations with respect to origin are additive.

Successive Scaling

When two successive scaling applied on object by two scaling factors: $S' : (S_{x'}, S_{y'})$ and $S'' : (S_{x''}, S_{y''})$ apply to a 2D object one by one known as successive scaling[12]. As shown in Fig. 5 initially ABCD is original position of object after first scaling S' the object is scaled to $A'B'C'D'$ positions and later on next scaling S'' takes place and then scaled object again scaled to the $A''B''C''D''$ position.

Scaling factors: $S' = (S_{x'}, S_{y'})$ and $S'' = (S_{x''}, S_{y''})$

Original object points are $A(x_1, y_1), B(x_2, y_2), C(x_3, y_3), D(x_4, y_4)$

After first scaling object points are $A'(x_1', y_1'), B'(x_2', y_2'), C'(x_3', y_3'), D'(x_4', y_4')$

After second scaling object points are $A''(x_1'', y_1''), B''(x_2'', y_2''), C''(x_3'', y_3''), D''(x_4'', y_4'')$

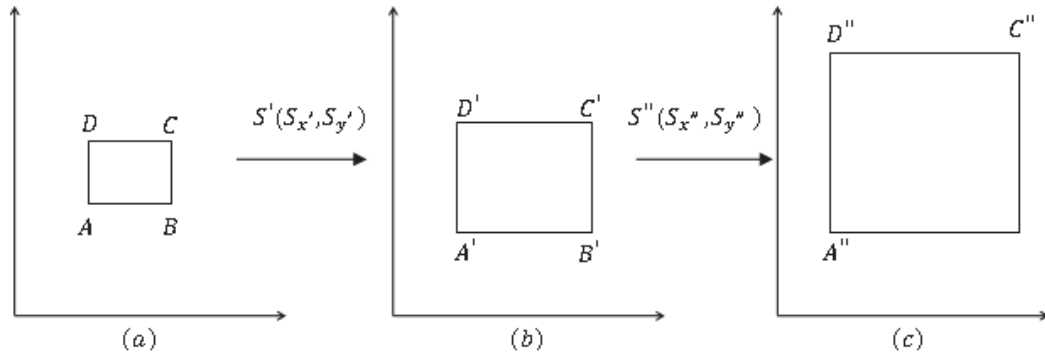


FIGURE 5. Successive scaling

Mathematical Equations and Matrix

Equation 12, 13 and 14 show the mathematical notation of successive scaling. The scaled transformed formulation is initial scaling.

$$\begin{aligned}
 P' &= S' \cdot P \\
 P'(x', y') &= S'(S_{x'}, S_{y'}) \cdot P(x, y) \\
 \text{where } x' &= x \cdot S_{x'}, y' = y \cdot S_{y'}
 \end{aligned} \tag{12}$$

Matrix:

$$\begin{bmatrix} S_{x'} & 0 & 0 \\ 0 & S_{y'} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Next scaling:

$$\begin{aligned}
 P'' &= S'' \cdot P' \\
 P''(x'', y'') &= S''(S_{x''}, S_{y''}) \cdot P'(x', y') \\
 \text{where } x'' &= x' \cdot S_{x''}, y'' = y' \cdot S_{y''}
 \end{aligned} \tag{13}$$

Matrix:

$$\begin{bmatrix} S_{x''} & 0 & 0 \\ 0 & S_{y''} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The transformation is scaled by two successive scaling operations of two dimensions produce the following composite scaling matrix.

$$\begin{aligned}
 P'' &= S'' \cdot S' \cdot P \\
 P''(x'', y'') &= S''(S_{x''}, S_{y''}) \cdot S'(S_{x'}, S_{y'}) \cdot P(x, y) \\
 \text{where } x'' &= x \cdot S_{x'} \cdot S_{x''}, y'' = y \cdot S_{y'} \cdot S_{y''} \\
 S &= S''(S_{x''}, S_{y''}) \cdot S'(S_{x'}, S_{y'})
 \end{aligned} \tag{14}$$

$$S = \begin{bmatrix} S_{x''} & 0 & 0 \\ 0 & S_{y''} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} S_{x'} & 0 & 0 \\ 0 & S_{y'} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$S = \begin{bmatrix} S_{x'} \cdot S_{x''} & 0 & 0 \\ 0 & S_{y'} \cdot S_{y''} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Both $S' = (S_{x'}, S_{y'})$ and $S'' = (S_{x''}, S_{y''})$ scaling are equal to the single scaling $S = ((S_{x'} \cdot S_{x''}), (S_{y'} \cdot S_{y''}))$. This demonstrates that two successive scaling with respect to origin are multiplicative.

DISCUSSION

With the help of matrix representation, settle down the sequence of transformation as composite matrix for transformation. The product of transformation matrices is same as referred to composition or concatenation of matrices. In the discussion of 2D successive translations are additive, multiplying the two rotation matrices discussion says that 2D successive rotation angles are additive and the resultant of 2D successive scaling are multiplicative. When matrix multiplication is going to perform then needs to consider some below properties[4].

1. As multiplication of matrix is associative, for example MATRIX_1, MATRIX_2 and MATRIX_3 three matrices are there then the matrix product can be like MATRIX_3.MATRIX_2.MATRIX_1. Associate property represented as $MATRIX_3.MATRIX_2.MATRIX_1 = (MATRIX_3.MATRIX_2).MATRIX_1 = MATRIX_3.(MATRIX_2.MATRIX_1)$. Composite matrix constructed by multiplying from right to left also known as post-multiplying and left to right also known as pre-multiplying.
2. As multiplication of matrix may not be commutative, for example if there are two matrices MATRIX_1, MATRIX_2 then MATRIX_1.MATRIX_2 is not equal to MATRIX_2.MATRIX_1 in general. When rotation and translation operations performed on an object then be careful with order. Otherwise when two operations are same such as two successive translation, rotation and scaling then multiplication of matrix is commutative.

CONCLUSION

Two dimensional (2D) successive translation, rotation and scaling are very useful in computer graphics. These compositions of transformation can apply in different areas such as image processing, medical science, gaming process and many more. Mathematical and matrix representation both are useful for solving the problems of 2D transformations. Mathematical and matrix notation can be helpful to the computer scientists in graphic evaluation through modification and enhancing the result of the output. In future general fixed point rotation and scaling can also represent in the form of matrix, mathematical formula can also take into consideration. Same as two dimensional, three dimensional successive transformations can be future aspect for researchers.

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